Estimating the cross-linguistic prevalence of the Sonority Sequencing Principle in adult and child-directed speech

Syllables are a privileged linguistic unit in speech processing during the earliest stages of language acquisition (Bertoncini et al., 1988; Eimas, 1999; Jusczyk & Derrah, 1987). For sintance, during as early as their first days of life, humans are able to discriminate between word-forms based on the number of syllables, instead of relying on the number of phonemes or the duration of the produced utterance (Bijeljac-Babic et al., 1993). Newborns can also track the co-ocurrence of syllable-sized groups of speech sounds to chunk the continuous speech stream into word-like units (Fló et al., 2019; 2022), an ability that will later give them access to the precursors of a lexicon (Saffran et al., 1996; Saffran & Kirkham, 2018). Given the relatively limited experience newborns have accumulated with their native language,their ability to identify syllables in speech is remarkable. To date, the mechanism behind such an early speech processing skill are unclear.

Previous work points to newborns being sensitive to certain constraints to syllabic structure (Bertoncini & Mehler, 1981; Gomez et al., 2014; Moon & Fifer, 1990). One of them is the Sonority Sequencing Principle (SSP). The SSP constrains the order of the segments inside a syllable based a phonological property name sonority, which indexes the perceptual salience of an individual segment relative to neighbouring ones (Albert & Nicenboim, 2022; Jany et al., 2007, for a discussion on the specific acoustic correlates of sonority; Parker, 2002; 2008). Sonority is conceived as an ordinal feature of individual speech sound (e.g., Clements (1990) suggests a hierarchy such as *Obstruents* < *Nasals* < *Liquids* < *Glides* < *Vowels*), which is oftentimes represented as a numerical scale. For methodological purposes, we will adopt the numerical scale proposed by Parker (2008), ranging from 1 (lowest sonority) to 8 (highest sonority). The SSP states that each syllable displays a unique peak of sonority in the nucleus, and that sonority rises monotonically from the syllable margins toward the nucleus (Parker, 2002; Selkirk, 1984). Following a numeric scale of sonority, a syllable can be evaluated as following or violating the SSP by examining its sonority contour, i.e., the sequence of sonority values assigned to each segment in the syllable (Blevins, 1995; Clements, 1990). Figure 1 illustrates the case of two English word-forms that follow and violate the SSP. The English word “crisp” /kɹɪsp/ complies with the SSP, with stop consonants /k/ and /p/ (lowest sonority) at the periphery of the syllable, /ɹ/ and /s/ (medium sonority) at surrounding the syllabic nucleus, and /ɪ/ (highest sonority) at the centre of the syllable. A word like “skirt” /skɪɾt/, on the other hand, violates the SSP, as its sonority contour does not increase monotonically towards the syllabic nucleus.

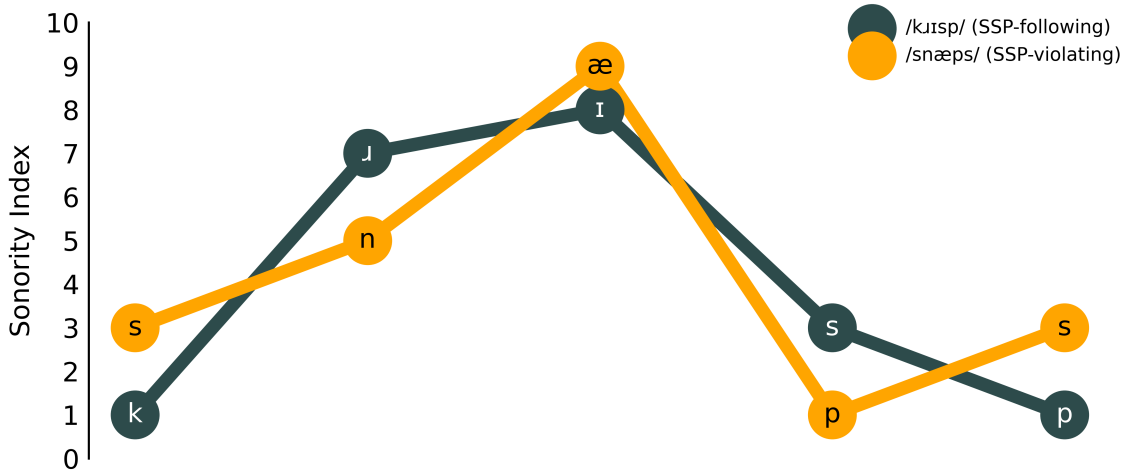


Figure 1: Examples of SSP-following (English word "crisp", /kɹɪsp/) and SSP-violating (English word "snaps", /snæps/) syllables according to the Sonority Index.

Evidence from experimental studies suggests that sonority—and particularly the SSP—play an important role in language processing and production (Berent et al., 2007; 2011; Berent & Lennertz, 2010; Chen et al., 2024; Miozzo & Buchwald, 2013), also in sign languages (Perlmutter, 1993; Sandler, 1993; Stone et al., 2018). regarding the role of the SSP in language acquisition, Gomez et al. (2014) showed in newborns that hearing SSP-following syllables (e.g., *bran*) elicits different brain haemodynamic responses compared to SSP-violating syllables (e.g., *lbif*). This suggests that newborns, who have not yet accumulated extensive experience with the phonological contraints of their native language, are already sensitive to violations of the SSP. Such early bias toward certain syllabic structures would provide humans with an available mechanism to detect syllables in the continuous speech stream at the earliest stages of language acquisition.

The SSP has been long claimed to be a linguistic universal, on the grounds of SSP-following syllables being preferred over SSP-violating syllables across most languages (Greenberg, 1978; Kreitman, 2012). For instance, Parker (2017) collected evidence from 275 experimental works on a typologically diverse list of 75 distinct languages, and found that 63% of the studies suggest that sonority plays a relevant role for accounting for the results, 22% provide inconclusive evidence, and 15% provide negative evidence. The universality of the SSP would place this principle as one of the few cues for speech segmentation present from birth to virtually *all* humans, before language-specific constraints to sequences of speech sounds (e.g., phonotactic rules) are available. Experimental evidence also supports that sonority constrictions are part of some language-non-specific knowledge that humans possess: listeners prefer SSP-following over SSP-violating consonant clusters, even when both types of structure are unattested in the native language (Berent et al., 2011; Berent & Lennertz, 2010).

The status of the SSP as a linguistic universal is nonetheless still a matter of debate, as violations of this principle are common across many languages. For instance, Yin et al. (2023) reported at least one violation of the SSP in around half of the 496 languages they examined. For instance, onset consonant clusters of the type initial fricative-stop consonant (e.g., /st/, as in "Stein" in German)

or its inverse counterpart in coda stop consonant-fricative (e.g., /ks/, as in “clicks” in English) are attested in many languages (Kawasaki-Fukumori et al., 1997). Some languages are known for allowing fairly more complex consonant clusters at syllabic onset whic violate the SSP. This is the case of Russian, which allows for clusters up to three consonants at syllabic onset (e.g., “взгляд” /vzglyad/ [glance]) (Halle & Jones, 1971; Itô, 1982) or Georgian, in which onset consonant clusters like /md/ are attested (e.g., “მდინარე” /ˈmdi.nä.ɾi/) (Crouch et al., 2023). Recent work on previously neglected languages also suggests to the proportion of SSP-violating consonant clusters may be higher than prior work had anticipated in some languages (Al Tamimi & Al Shboul, 2013, in Modern Standard Arabic; Sun & Billington, 2024, in Lelepa). In light of these findings, the SSP cannot be taken as a hard linguisitic universal, but rather as an overall preference toward certain linguistic structures that that virtually all languages display to a varying extent (Frisch, 2015; Prince & Smolensky, 2004). Under this account, the role of the SSP as a universally available cue for syllabification becomes a quantitative question: how often do syllables in different languages abide by the SSP?

The aim of the present paper is to provide empirical estimates of the prevalence of the SSP in a fairly representative sample of 139 languages from 17 distinct language families. We pursued this goal in two ways. In a first analysis, we estimated the proportion of the syllabic repertoire of each language that followed SSP. Using Bayesian modelling we derived a quantitative estimate of global and language-specific SSP prevalence, along with their corresponding statistical uncertainty. Because items inside the syllabic repertoire of a language likely occur at different rates in speech (e.g., both /ka/ and /gla/ syllables are attested in Spanish, but the former likely appears in a larger number of words), a second analysis investigated whether SSP-following syllables are also more frequent speech compared to SSP-violating syllables. We explored this research question in two datasets. In the first dataset, syllabic frequencies were derived from available databases of lexical frequencies that did not discriminate between adult-directed speech (ADS) and child-directed speech (CDS). In the second dataset, syllabic frecuencies were derived from lexical frequencies in the CHILDES (MacWhinney & Snow, 1990), which includes corpora of child-directed speech.

1 Analysis 1: What proportion of the syllabic repertoire of a language follows the SSP?

1.1 Methods

Raw data, reproducible code, and project documentation are available in the Open Science Framework (https://osf.io/yeqms) and GitHub (https://github.com/NeuroDevCo/ssp-corpus).

1.1.a Data collection

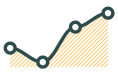
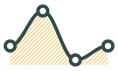
We scraped all phonological transcriptions from word entries in the Wiktionary database. Wiktionary is an open-source, free database of lexical entries. As of June 10th 2026, Wiktionary contains 10,050,751 lexical entries from 8,244 langages contributed by 4,463,976 total users. Some lexical entries in Wiktionary also include a phonological in IPA format or phonetic transcription of the word-form in a particular language. We used the `wikipron` Python package (Lee et al., 2020) to scrape all available phonological transcriptions from all 7859 languages included in the

Glottolog database (Nordhoff & Hammarström, 2011). This resulted in ~1.7 million pronunciations belonging to 900 languages. We filtered out languages with less than 100 phonological transcriptions available. The final dataset containing word-form phonological transcriptions included 731,145 word-forms from 192 languages. Information from each language was extracted using the `lingtypology` Python package (Moroz, 2017; Voronov & Forkel, 2019).

1.1.b Data processing

We segmented word-form phonological transcriptions into syllable candidates where possible. If available, we used explicit syllable boundary markers (`.`) or implicity syllable markers (`ˈ` for primary stress, `ˌ` for secondary stress). In the absence of syllabic markers, we assumed the word to be monosyllabic. We then classified the segments inside each resulting syllable candidate into consonants (`C`), vowels (`V`), and semi-vowels (`v`). Consecutive sequencies of vowels and semi-vowels (e.g., `V`, `VV`, `Vv`, `VVv`) where classified as vocalic clusters. syllable candidates with more than one vocalic cluster were flagged as multisyllabic and excluded from the dataset. We also remove duplicated syllables within each language (e.g., if the syllable /ka/ appeared in Spanish words /ˈka.sa/ and /kaˈpo/ in Spanish, it would appear more than once in the Spanish dataset). The final result was a syllabic repertoire for each language. To evaluate if each extracted syllable followed or violated the SSP, we assigned a sonority score to each segment in each syllable in a scale from 1 to 9, according to the `panphon` Python package (Mortensen et al., 2016). Syllables with more than one peak (i.e., local minimum) in the sonority sequence were classified as SSP-violating, and as SSP-following otherwise. Syllables with a flat sonority contour were classifid as SSP-following.

Table 1: Examples of SSP-following and SSP-violating syllables in the dataset. Sonority index contours are shown for each syllable numerically and graphically.

Language	Syllable	Original word	Sonority index	
SSP-following				
Esperanto	/aj̯/	aj /aj̯/	9-7	
Hausa	/bàb/	babba /bàb.bá/	2-9-2	
Eastern Armenian	/bed/	կարպետ /gɑrˈbed/	2-9-2	
Kaingang	/han/	han /han/	6-9-5	
Scots	/ald/	auld /ald/	9-6-2	
SSP-violating				
Emiliano	/skri/	scritrîz /skriˈtriːθ/	3-1-6-8	
Modern Hebrew	/jruː/	ירושלים /jruːˌΣɔːˈlaˌjim/	7-6-8	
Hunsrik	/fiks/	fix /ˈfiks/	3-8-1-3	
Javanese	/ŋsɔ/	ꦧꦺꦴ /bɔ.ŋsɔ/	5-3-9	
Old Russian	/jsʇ͡ɕi/	исчисти /jsʇ͡ɕiˈstʲi/	7-3-1-8	

1.1.c Data analysis

To model the prevalence of the SSP across languages, we aggregated, for each language in the dataset, the number of syllables that followed the SSP. This resulted in one data point per language. We used a multilevel, Bayesian logistic regression model to estimate the proportion of syllables that followed or violated the SSP across languages. This model estimated the average probability of the syllables of each language in the dataset following the SSP (θ_j). This model included each family language as a group with a random intercept, and a bias correction parameter that adjusted θ_j by number of observations in each language's dataset (see Section 4.2 for more

details). Equation 1 shows a detailed implementation of both models model. We implemented the model in the probabilistic programming language Stan (Carpenter et al., 2017) and CmdStanPy, its interface to its Python 3.12.7. We run eight Hamiltonian Monte Carlo chain of 2,000 iterations each (1,000 of which were warm-up iterations) to estimate the posterior distribution of the model.

Likelihood:

$$y_j \sim \text{Binomial } (n_j, \frac{\theta_j}{1-\theta_j})$$

$$\theta_j = \mu_j + u_j + (\gamma / n_j)$$

where:

i represents each observation, corresponding to a language
y_j is the number of syllables in language i that follow SSP
u_j is a language-wise adjustment
n_i is the number of syllables total language i
γ introduces an adjustment to μ that is inverselly proportional to n_i

(1)

Prior:

$$\mu_j \sim \text{Normal } (0, 5)$$

$$\sigma_u \sim \text{Exponential } (4)$$

1.2 Results

The final dataset included 113,760 syllables from 139 languages from 17 distinct language families. Table 3 shows the number of SSP-following and SSP-violating syllables for each language family in the dataset (see Section 4.1 for a more details). The median of the posterior distribution of the cross-language proportion of SSP-following syllables was θ = 0.981 (95% *HDI* = [0.975, 0.985]), and the median of the posterior distribution of the bias parameter was γ = 0.069 (95% HDI = [−1.556, 1.666]) (see Figure 1). Figure 3 shows the posterior mean and 95% HDI of the proportion of SSP-following syllables for each language family in the dataset.

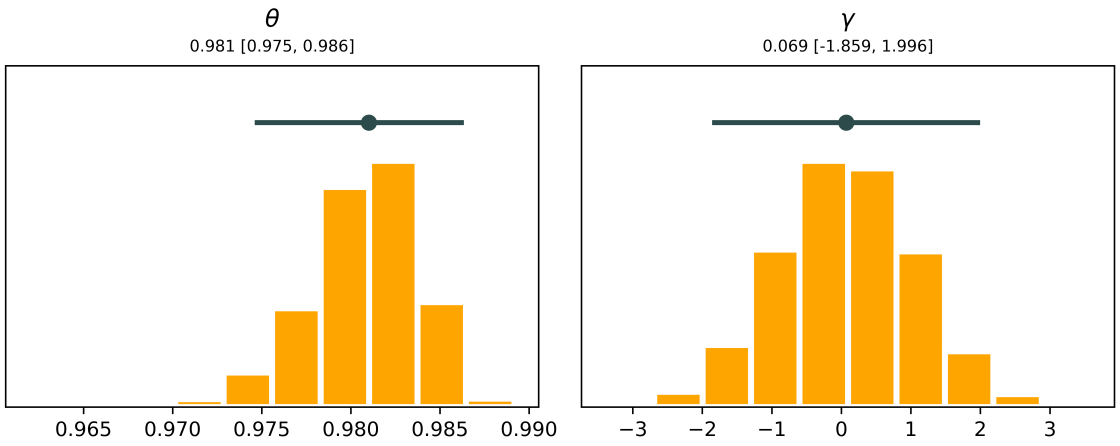


Figure 2: Posterior distribution of parameters of interest θ (cross-language prevalence of SSP-following syllables) and γ (bias parameter) in Equation 1

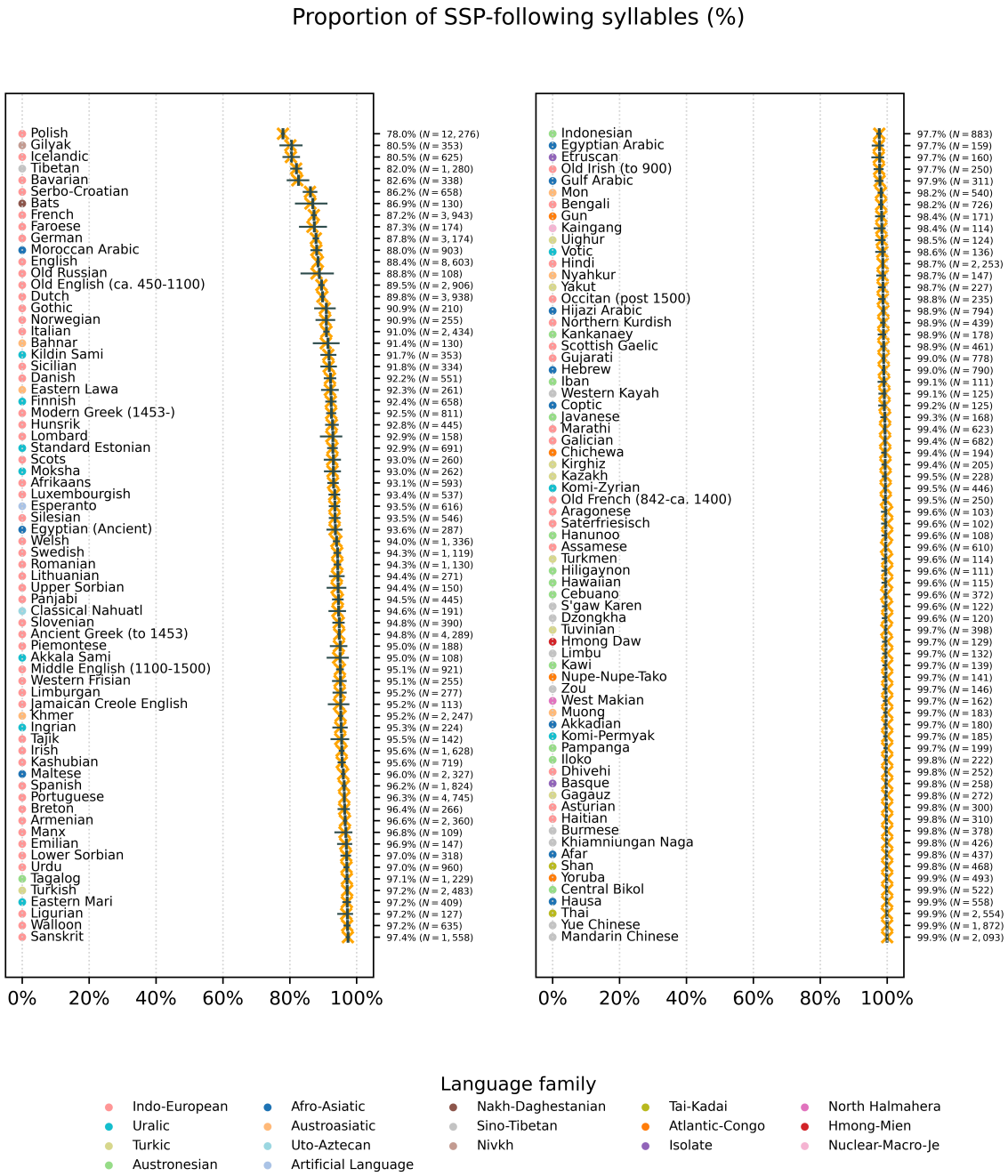


Figure 3: Posterior model-estimated proportion of distinct syllables that follow the Sonority Sequencing Principle (SSP) for each language family. The X-axis indicates the proportion of syllables that follow the SSP. The Y-axis indicates each language family. Black diamonds and intervals indicate the posterior mean proportion of SSP-following syllables for each language family, and its 95% highest density interval (*HDI*). Means and 95% *HDI* are also numerically reported next to each language family.

2 Analysis 2: Are SSP-following syllables more frequent than SSP-violating syllables?

2.1 Method

2.1.a Data collection

2.1.a.a Extracting lexical frequencies

We extracted adult lexical frequencies for each word-form in the words dataset using the `wordfreq` Python package (Speer, 2022). This package provides an interface to retrieve lexical frequency data for words in over 40 languages. Lexical frequencies are extracted from a variety of sources according to their availability for each language, including Wikipedia, subtitles (e.g., OPUS Open subtitles 2018 and SUBTLEX), news, books, web text, Twitter, and Reddit, among others. At least three of these sources contribute to the calculation of lexical frequencies in each of the available languages[1]. When lexical frequencies were not available for a particular language (e.g., the ISO 639-3 code does not match any available language in `wordfreq`), an equivalent language was used, if available. Equivalent languages were searched according to their distance to the original language according to CLDR v38.1. This system assigns a numeric value to each pair of language codes based on whether both codes are assigned to identical languages with different names (e.g., Norwegian Bokmål → Norwegian, score 0), minor regional differences (score 1-3, e.g., Norwegian Bokmål → Norwegian), and languages with significant but unproblematic regional differences (score 10-24, e.g., Afrikaans → Dutch, Wu Chinese → Mandarin Chinese), etc. The rationale behind using equivalent languages to retrieve missing lexical frequencies was that users of close languages are likely to use similar words to refer to the same objects, and therefore lexical frequencies are expected to be reasonably equivalent across such language pairs.

2.1.a.b Calculating syllabic frequencies

For each syllable, we aggregated the lexical frequency (frequency per million words) of each word the syllable was part of in the dataset. The rationale behind this step is that each time a word is uttered, each of its syllables is also uttered. Therefore, by calculating the lexical frequency of each word a given syllable is involved in, an estimation of syllabic frequency can be derived. Table 2 provides illustrative examples of how syllabic frequencies were calculated.

[1]Importantly, lexical frequencies provided by `wordfreq` have have not been updated since September 2024 to prevent AI generated text from biasing the data.

Table 2: Illustrative example of syllabic frequency calculation for syllables /ka/ and /ne/ across French and Spanish. *PMW*: Frequency per million words. *Zipf*: Zipf-transformed frequency score.

	French			Spanish		
	Word	Generic	CHILDES	Word	Generic	CHILDES
Syllable /ka/						
	/ka.do/	44.67	193.55	/kaˈido/	18.62	522.87
	/ɛ̃.ka.pabl/	19.50	0.61	/kaˈsiʒa/	5.62	4.88
	/ʃi.ka.ɡo/	15.85	1.22	/kaˈmat͡ʃo/	4.79	0.44
	/ka.naʁ/	14.79	255.73	/kaˈbeɾ/	1.51	27.08
	/si.ka.tʁis/	2.63	4.57	/kaˈbaʃa/	0.37	0.89
PMW		97.44	455.68		30.92	556.16
Zipf		4.99	5.66		4.49	5.75
Syllable /ne/						
	/ʁə.kɔ.ne.s̃s̃/	36.31	0.30	/neˈɡoθjo/	85.11	1.78
	/ʁə.ne/	20.89	3.66	/neˈgɾita/	2.40	4.44
	/ʃə.mi.ne/	7.08	61.57	/kaɾˈne/	2.00	0.89
	/sta.sjɔ.ne/	1.10	0.91	/neˈbados/	0.63	2.22
	/ʃ.ki.ki.ne/	0.13	0.30	/neˈnito/	0.24	0.89
PMW		65.51	66.75		90.38	10.21
Zipf		4.82	4.82		4.96	4.01

2.1.b Data analysis

We used a multilevel, Bayesian, distributional regression model to estimate the proportion of syllables that followed or violated the SSP across languages. The response variable (Frequency) was modelled using a log-normal distribution, with a location parameter μ and a scale parameter σ. We modelled both parameters using linear regression models, each with a global intercept (β_0, π_0) with a language-specific adjustment (u_{0j}, w_{0j}), a predictor variable SSP, which indexed whether the syllable followed the SSP (coded as `+0.5`) or violated the SSP (coded as `-0.5`) and its corresponding regression coefficient (β_1, π_1), with language-specific adjustments (u_{1j}, w_{1j}). Equation 2 provides a formal definition of the model. We specified weakly informative priors for the parrameters in the model. We run eight Hamiltonian Monte Carlo chain of 2,000 iterations each (1,000 of which were warm-up iterations) to estimate the posterior distribution of the model.

Likelihood:

$$
\begin{aligned}
y_{ij} &\sim \text{LogNormal } (\mu_{ij}, \sigma_{ij}) \\
\mu_{ij} &= \beta_0 + u_{0j} + (\beta_1 + u_{1j}) \cdot \text{SSP}_{ij} \\
\sigma_{ij} &= \exp(\pi_0 + w_{0j} + (\pi_1 + w_{1j}) \cdot \text{SSP}_{ij})
\end{aligned}
$$

Prior:

$$
\begin{aligned}
\beta_{[0,1]} &\sim \text{Normal } (0, 10) \\
\pi_{[0,1]} &\sim \text{Normal } (0, 5) \\
u_{[0,1]j} &\sim \text{Normal } (0, \tau_{uj}) \\
w_{[0,1]j} &\sim \text{Normal } (0, \tau_{wj}) \\
\tau_{uj}, \tau_{wj} &\sim \text{Exponential } (2)
\end{aligned}
\tag{2}
$$

2.2 Results

The final dataset for adult lexical frequencies comprised 52,588 syllables from 42 languages from 8 distinct language families. The results from the Bayesian model indicate that the 95% HDI of the difference between SSP-following and SSP-violating syllables in the location parameter did not exceed zero (μ = 0.024, 95% *HDI* = [0.003, 0.044]), suggesting no evidence that SSP-following syllables are more frequent than SSP-violating syllables across languages.

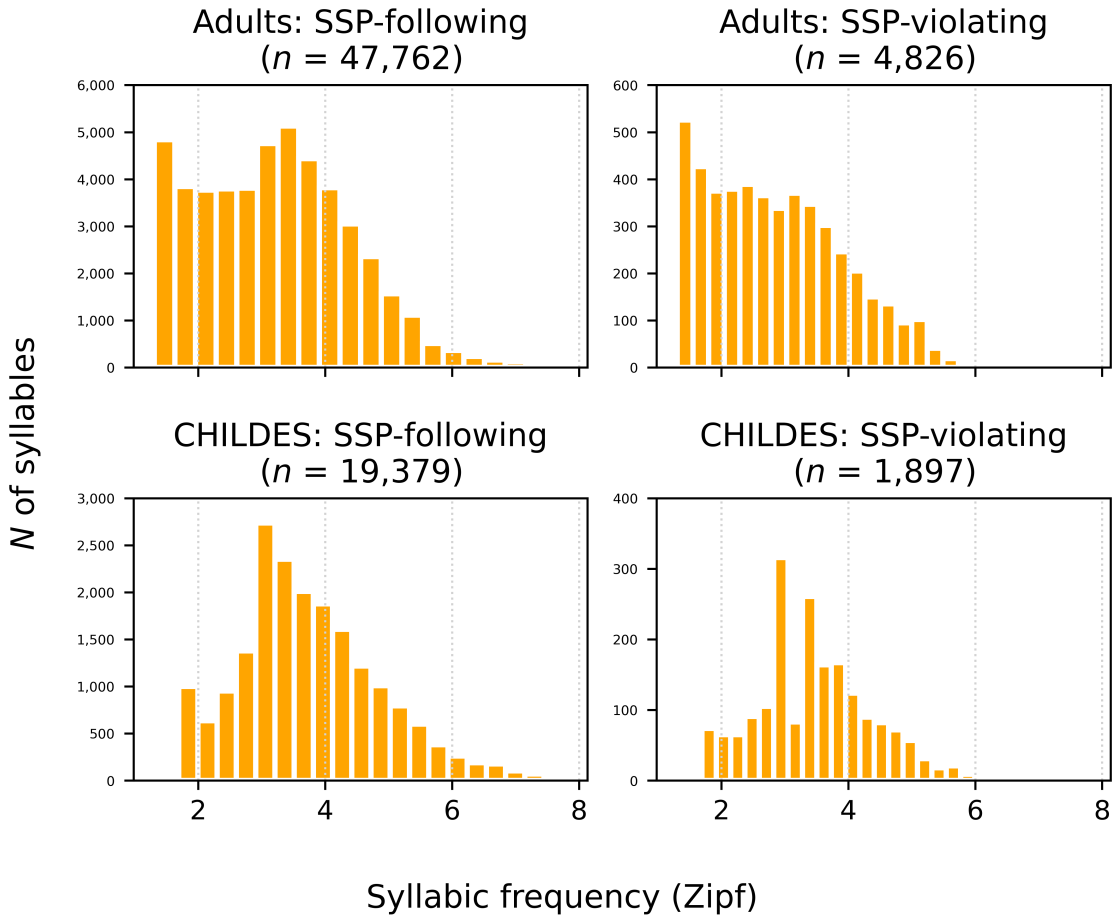
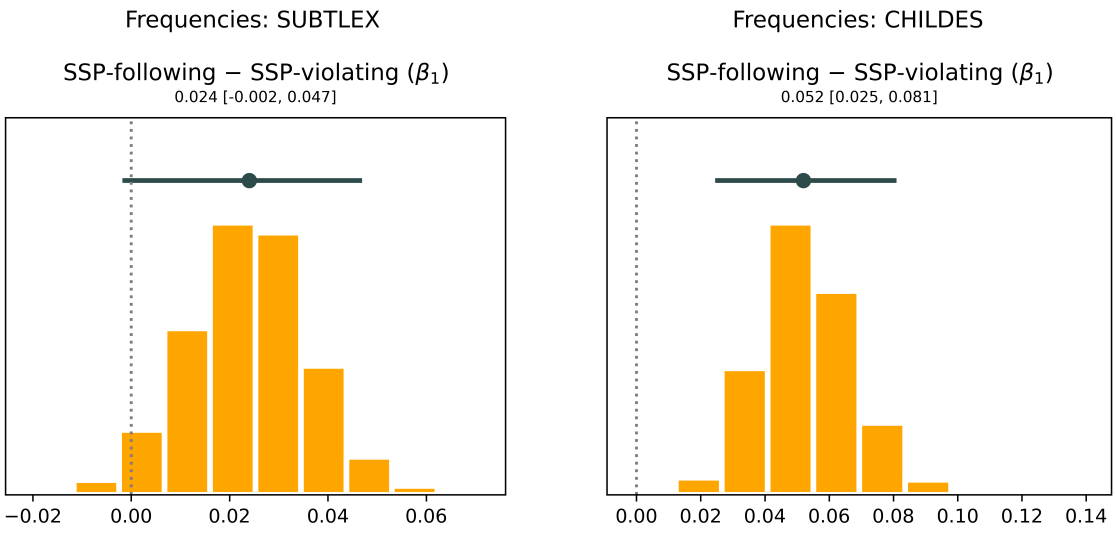


Figure 4: Distribution of syllabic frequencies across languages for SSP-followingg and SSP-violating syllables.

For lexical frequencies extracted from CHILDES, the number of syllables in the dataset was 21,276 from 17 languages from 3 distinct language families. In this dataset, the posterior distribution of the difference between SSP-following and SSP-violating syllables in the location parameter did exceed zero (μ = 0.052, 95% *HDI* [0.030, 0.078]), indicating that SSP-following syllables are more frequent than SSP-violating syllables across languages.

Figure 4 shows the distribution of syllabic frequencies across languages for SSP-following and SSP-violating syllables. The posterior distribution of the model parameters is shown in Figure 5.



(a) Adult lexical frequencies (b) Infant lexical frequencies (CHILDES)

Figure 5: Posterior distribution of main parameters in Equation 2.

3 Discussion

4 Appendix

4.1 Number of SSP-following and SSP-violating syllables per language and language family

Table 3: Number of SSP-following and SSP-violating syllables per language and language family.

	N syllables	N words	Following	Violating	% Following
Afro-Asiatic					
Moroccan Arabic	903	833	794	109	87.93%
Egyptian (Ancient)	287	942	268	19	93.38%
Maltese	2,327	3,777	2,233	94	95.96%
Egyptian Arabic	159	148	155	4	97.48%
Gulf Arabic	311	343	304	7	97.75%
Hijazi Arabic	794	952	785	9	98.87%
Modern Hebrew	790	948	782	8	98.99%
Coptic	125	128	124	1	99.20%
Akkadian	180	184	180	0	100.00%

	N syllables	N words	Following	Violating	% Following
Afar	437	1,335	437	0	100.00%
Hausa	558	738	558	0	100.00%
Artificial Language					
Esperanto	616	5,975	575	41	93.34%
Atlantic-Congo					
Gun	171	143	168	3	98.25%
Nyanja	194	197	193	1	99.48%
Nupe-Nupe-Tako	141	153	141	0	100.00%
Yoruba	493	1,311	493	0	100.00%
Austroasiatic					
Bahnar	130	198	118	12	90.77%
Eastern Lawa	261	254	240	21	91.95%
Central Khmer	2,247	6,070	2,139	108	95.19%
Mon	540	1,051	530	10	98.15%
Nyahkur	147	152	145	2	98.64%
Muong	183	195	183	0	100.00%
Austronesian					
Tagalog	1,229	13,663	1,193	36	97.07%
Standard Indonesian	883	3,851	862	21	97.62%
Kankanaey	178	422	176	2	98.88%
Iban	111	254	110	1	99.10%
Javanese	168	145	167	1	99.40%
Cebuano	372	1,556	371	1	99.73%
Kawi	139	218	139	0	100.00%
Hawaiian	115	503	115	0	100.00%
Hanunoo	108	239	108	0	100.00%
Coastal-Naga Bikol	522	2,165	522	0	100.00%

	N syllables	N words	Following	Violating	% Following
Hiligaynon	111	230	111	0	100.00%
Pampanga	199	313	199	0	100.00%
Iloko	222	510	222	0	100.00%
Hmong-Mien					
Hmong Daw	129	124	129	0	100.00%
Indo-European					
Polish	12,276	123,564	9,570	2,706	77.96%
Icelandic	625	2,346	502	123	80.32%
Bavarian	338	756	278	60	82.25%
Serbian-Croat-ian-Bosnian	658	8,822	566	92	86.02%
Faroese	174	563	151	23	86.78%
French	3,943	17,659	3,438	505	87.19%
German	3,174	13,142	2,786	388	87.78%
Old Russian	108	233	95	13	87.96%
English	8,603	36,929	7,605	998	88.40%
Old English (ca. 450-1100)	2,906	12,302	2,599	307	89.44%
Dutch	3,938	11,340	3,535	403	89.77%
Gothic	210	404	190	20	90.48%
Norwegian	255	451	231	24	90.59%
Italian	2,434	15,211	2,214	220	90.96%
Sicilian	334	519	306	28	91.62%
Danish	551	1,399	507	44	92.01%
Modern Greek	811	3,104	749	62	92.36%
Lombard	158	199	146	12	92.41%
Hunsrik	445	888	412	33	92.58%
Scots	260	370	241	19	92.69%
Afrikaans	593	619	551	42	92.92%

	N syllables	N words	Following	Violating	% Following
Moselle Franconian	537	860	501	36	93.30%
Silesian	546	518	510	36	93.41%
Welsh	1,336	5,150	1,255	81	93.94%
Upper Sorbian	150	372	141	9	94.00%
Lithuanian	271	279	255	16	94.10%
Swedish	1,119	1,624	1,054	65	94.19%
Romanian	1,130	2,126	1,065	65	94.25%
Eastern Panjabi	445	452	420	25	94.38%
Slovenian	390	1,704	369	21	94.62%
Piemontese	188	264	178	10	94.68%
Jamaican Creole English	113	150	107	6	94.69%
Ionic-Attic Ancient Greek	4,289	28,165	4,065	224	94.78%
Western Frisian	255	425	242	13	94.90%
Limburgan	277	205	263	14	94.95%
Middle English	921	2,827	875	46	95.01%
Tajik	142	258	135	7	95.07%
Irish	1,628	4,682	1,555	73	95.52%
Kashubian	719	689	687	32	95.55%
Spanish	1,824	106,961	1,755	69	96.22%
Breton	266	286	256	10	96.24%
Portuguese	4,745	99,468	4,569	176	96.29%
Manx	109	186	105	4	96.33%
Eastern Armenian	2,360	5,906	2,278	82	96.53%
Emiliano	147	167	142	5	96.60%
Ligurian	127	194	123	4	96.85%

	N syllables	N words	Following	Violating	% Following
Lower Sorbian	318	647	308	10	96.86%
Urdu	960	1,088	931	29	96.98%
Walloon	635	612	617	18	97.17%
Sanskrit	1,558	2,831	1,517	41	97.37%
Early Irish	250	392	244	6	97.60%
Bengali	726	2,771	713	13	98.21%
Hindi	2,253	5,072	2,223	30	98.67%
Occitan	235	243	232	3	98.72%
Northern Kurdish	439	508	434	5	98.86%
Scottish Gaelic	461	1,112	456	5	98.92%
Gujarati	778	870	770	8	98.97%
Marathi	623	703	619	4	99.36%
Galician	682	3,477	678	4	99.41%
Old French (842-ca. 1400)	250	295	249	1	99.60%
Assamese	610	1,379	608	2	99.67%
Aragonese	103	178	103	0	100.00%
Asturian-Leonese-Cantabrian	300	1,181	300	0	100.00%
Haitian	310	848	310	0	100.00%
Dhivehi	252	326	252	0	100.00%
Ems-Weser Frisian	102	163	102	0	100.00%
Isolate					
Etruscan	160	206	156	4	97.50%
Basque	258	10,555	258	0	100.00%
Nakh-Daghestanian					
Bats	130	253	112	18	86.15%

	N syllables	N words	Following	Violating	% Following
Nivkh					
Amur Nivkh	353	569	283	70	80.17%
North Halmahera					
West Makian	162	325	162	0	100.00%
Nuclear-Macro-Je					
Kaingang	114	107	112	2	98.25%
Sino-Tibetan					
Tibetan	1,280	1,208	1,048	232	81.88%
Western Kayah	125	129	124	1	99.20%
Limbu	132	107	132	0	100.00%
Burmese	378	2,430	378	0	100.00%
Mandarin Chinese	2,093	2,577	2,093	0	100.00%
S'gaw Karen	122	220	122	0	100.00%
Zou	146	145	146	0	100.00%
Dzongkha	120	104	120	0	100.00%
Khiamniungan Naga	426	2,952	426	0	100.00%
Yue Chinese	1,872	1,875	1,872	0	100.00%
Tai-Kadai					
Thai	2,554	3,564	2,552	2	99.92%
Shan	468	361	468	0	100.00%
Turkic					
Turkish	2,483	5,108	2,412	71	97.14%
Uighur	124	967	122	2	98.39%
Sakha	227	214	224	3	98.68%
Kirghiz	205	181	204	1	99.51%
Kazakh	228	195	227	1	99.56%
Tuvinian	398	313	397	1	99.75%

	N syllables	N words	Following	Violating	% Following
Turkmen	114	157	114	0	100.00%
Gagauz	272	311	272	0	100.00%
Uralic					
Kildin Saami	353	659	323	30	91.50%
Finnish	658	73,343	607	51	92.25%
Moksha	262	334	243	19	92.75%
Estonian	691	2,050	641	50	92.76%
Akkala Saami	108	180	102	6	94.44%
Ingrian	224	8,558	213	11	95.09%
Eastern Mari	409	659	397	12	97.07%
Votic	136	1,876	134	2	98.53%
Komi-Zyrian	446	908	444	2	99.55%
Komi-Permyak	185	229	185	0	100.00%
Uto-Aztecan					
Classical Nahuatl	191	865	180	11	94.24%

4.2 Relationship between number of observations and SSP prevalence

Because we did not have access to the full syllabic repertoire of each language, available syllables from languages are the result of convenience sampling (e.g., in Wiktionary word entries are created by volunteers). The number of entries from each language, the amount of detail with which they are entered, and the type of words that are entered, varies systematically across languages. This might bias the estimates. For instance, if SSP-violating syllables are less frequent than SSP-following syllables in the lexicon of a language, and less frequent words are less likely to enter Wiktionary by volunteers, then the observed prevalence of SSP for that language likely overestimates the prevalence of SSP-following languages for a similar discussion (Yin et al., 2023, for a similar observation). For this reason, this model included a term in the linear model γ/n_i aimed at capturing this potential overrepresentation of SSP-following syllables in datasets with fewer observations.

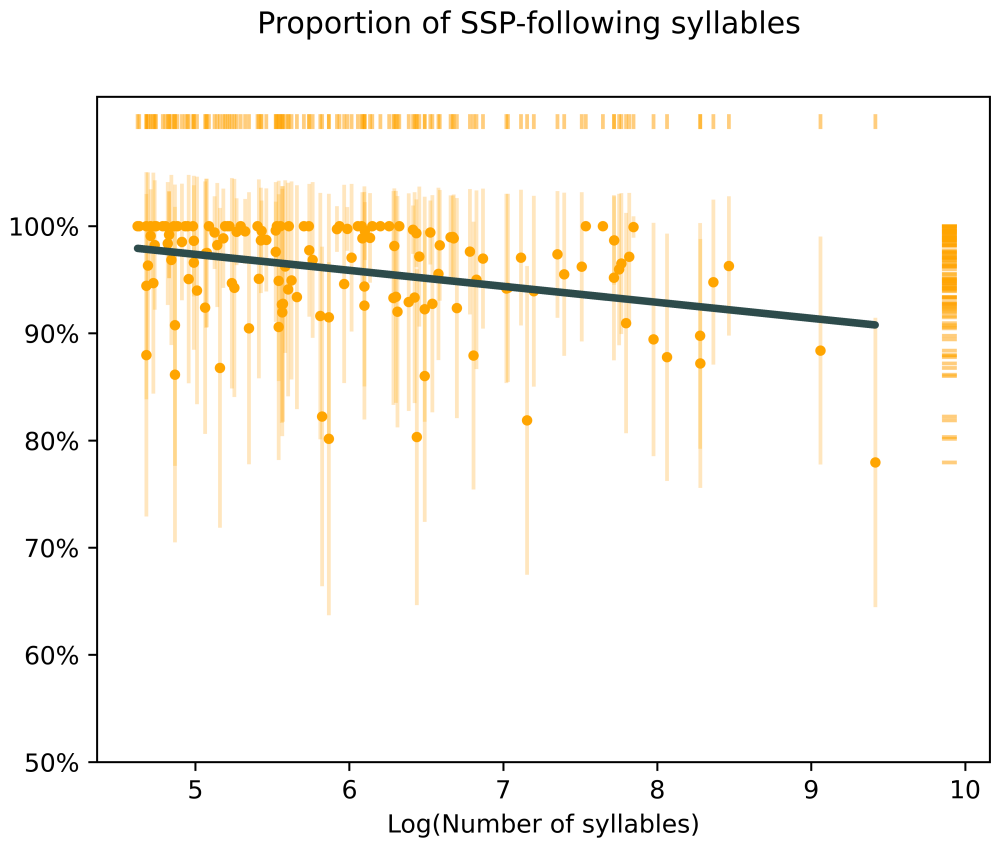


Figure 6: Relationship between the number of observed syllables and the SSP-prevalence in each language. The X-axis indicates the log-transformed number of observed syllables in the dataset. The Y-axis indicates the SSP prevalence (i.e., observed proportion of syllables that follow the SSP in the language's dataset). Each dot and interval indicate the data for one language, and the standard error of the SSP-prevalence. Marks in the top and right margin of the figure indicate the distribution of the observations. The predictions of a linear regression model between both variables are shown as a green line.

5 References

Bibliography

Al Tamimi, Y. A., & Al Shboul, Y. (2013). Is the phonotactics of the Arabic complex coda sonority-based?. *Journal of King Saud University-Languages and Translation*, *25*(1), 21–33.

Albert, A., & Nicenboim, B. (2022). Modeling Sonority in Terms of Pitch Intelligibility With the Nucleus Attraction Principle. *Cognitive Science*, *46*(7), e13161. https://doi.org/10.1111/cogs.13161

Berent, I., & Lennertz, T. (2010). Universal Constraints on the Sound Structure of Language: Phonological or Acoustic?. *Journal of Experimental Psychology: Human Perception and Performance*, *36*(1), 212–223. https://doi.org/10.1037/a0017638

Berent, I., Harder, K., & Lennertz, T. (2011). Phonological Universals in Early Childhood: Evidence from Sonority Restrictions. *Language Acquisition*, *18*(4), 281–293. https://doi.org/10.1080/10489223.2011.580676

Berent, I., Steriade, D., Lennertz, T., & Vaknin, V. (2007). What We Know about What We Have Never Heard: Evidence from Perceptual Illusions. *Cognition*, *104*(3), 591–630. https://doi.org/10.1016/j.cognition.2006.05.015

Bertoncini, J., & Mehler, J. (1981). Syllables as units in infant speech perception. *Infant Behavior and Development*, *4*, 247–260.

Bertoncini, J., Bijeljac-Babic, R., Jusczyk, P. W., Kennedy, L. J., & Mehler, J. (1988). An investigation of young infants' perceptual representations of speech sounds. *Journal of Experimental Psychology: General*, *117*(1), 21.

Bijeljac-Babic, R., Bertoncini, J., & Mehler, J. (1993). How Do 4-Day-Old Infants Categorize Multisyllabic Utterances?. *Developmental Psychology*, *29*(4), 711.

Blevins, J. (1995). The Syllable in Phonological Theory. In J. Goldsmith (Ed.), *Handbook of Phonological Theory: Vol. 1. Handbook of Phonological Theory* (pp. 206–244). Blackwell.

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., & Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, *76*, 1–32.

Chen, X., Ridouane, R., & Hallé, P. A. (2024). Sonority patterns and onset cluster production in Mandarin. *Proc. Issp 2024*, 8–11.

Clements, G. N. (1990). The Role of the Sonority Cycle in Core Syllabification. *Papers in Laboratory Phonology*, 283–333. https://doi.org/10.1017/cbo9780511627736.017

Crouch, C., Katsika, A., & Chitoran, I. (2023). Sonority sequencing and its relationship to articulatory timing in Georgian. *Journal of the International Phonetic Association*, *53*(3), 1049–1072.

Eimas, P. D. (1999). Segmental and syllabic representations in the perception of speech by young infants. *The Journal of the Acoustical Society of America*, *105*(3), 1901–1911.

Fló, A., Benjamin, L., Palu, M., & Dehaene-Lambertz, G. (2022). Sleeping neonates track transitional probabilities in speech but only retain the first syllable of words. *Scientific Reports*, *12*(1), 4391.

Fló, A., Brusini, P., Macagno, F., Nespor, M., Mehler, J., & Ferry, A. L. (2019). Newborns are sensitive to multiple cues for word segmentation in continuous speech. *Developmental Science*, *22*(4), e12802.

Frisch, S. A. (2015). A Preliminary Investigation of Quantitative Patterns in Sonority Sequencing. *Italian Journal of Linguistics*, *27*(1), 9–27.

Gomez, D., Berent, I., Benavides-Varela, S., Bion, R., Cattarossi, L., Nespor, M., & Mehler, J. (2014). Language Universals at Birth. *Proceedings of the National Academy of Sciences*, *111*(16), 5837–5841.

Greenberg, J. (1978,). *Some generalizations concerning initial and final consonant clusters. Universals of Human Language, eds Greenberg JH, Ferguson CA, Moravcsik EA*. Stanford Univ Press, Stanford.

Halle, M., & Jones, L. G. (1971). *The Sound Pattern of Russian: A Linguistic and Acoustical Investigation*. Walter de Gruyter.

Itô, J. (1982). The Syllable Structure of Russian. *Unpublished Ms. University of Massachusetts-Amherst*.

Jany, C., Gordon, M., Nash, C. M., & Takara, N. (2007). How Universal Is the Sonority Hierarchy?: A Cross-Linguistic Acoustic Study. *Proceedings of the ICPhS*, 1401–1404.

Jusczyk, P. W., & Derrah, C. (1987). Representation of speech sounds by young infants. *Developmental Psychology*, *23*(5), 648.

Kawasaki-Fukumori, H., Ohala, J., Eliasson, S., & Jahr, E. (1997). Alternatives to the Sonority Hierarchy for Explaining Segmental Sequential Constraints. *Language and Its Ecology: Essays in Memory of Einar Haugen*, *100*, 343.

Kreitman, R. (2012). On the relations between [sonorant] and [voice]. *Consonant Clusters and Structural Complexity*, *26*, 33.

Lee, J. L., Ashby, L. F., Garza, M. E., Lee-Sikka, Y., Miller, S., Wong, A., McCarthy, A. D., & Gorman, K. (2020). Massively multilingual pronunciation modeling with WikiPron. *Proceedings of the Twelfth Language Resources and Evaluation Conference*, 4223–4228.

MacWhinney, B., & Snow, C. (1990). The child language data exchange system: An update. *Journal of Child Language*, *17*(2), 457–472.

Miozzo, M., & Buchwald, A. (2013). On the nature of sonority in spoken word production: Evidence from neuropsychology. *Cognition*, *128*(3), 287–301.

Moon, C., & Fifer, W. P. (1990). Syllables as signals for 2-day-old infants. *Infant Behavior and Development*, *13*(3), 377–390.

Moroz, G. (2017). *lingtypology: easy mapping for Linguistic Typology*. https://cran.r-project.org/package=lingtypology

Mortensen, D. R., Littell, P., Bharadwaj, A., Goyal, K., Dyer, C., & Levin, L. (2016). Panphon: A resource for mapping IPA segments to articulatory feature vectors. *Proceedings of COLING 2016, The 26th International Conference on Computational Linguistics: Technical Papers*, 3475–3484.

Nordhoff, S., & Hammarström, H. (2011). Glottolog/Langdoc: Defining dialects, languages, and language families as collections of resources. In T. Kauppinen, L. C. Pouchard, & K. Keßler (Eds.), *Proceedings of the First International Workshop on Linked Science 2011 (LISC2011), Bonn, Germany, October 24, 2011: Proceedings of the First International Workshop on Linked Science 2011 (LISC2011), Bonn, Germany, October 24, 2011* (pp. 1–7). CEUR Workshop Proceedings. http://iswc2011.semanticweb.org/fileadmin/iswc/Papers/Workshops/LISC/nordhoff.pdf

Parker, S. G. (2002). *Quantifying the sonority hierarchy*. University of Massachusetts Amherst.

Parker, S. (2008). Sound Level Protrusions as Physical Correlates of Sonority. *Journal of Phonetics*, *36*(1), 55–90.

Parker, S. (2017). Sounding out Sonority. *Language and Linguistics Compass*, *11*(9), e12248. https://doi.org/10.1111/lnc3.12248

Perlmutter, D. M. (1993). Sonority and syllable structure in American Sign Language. In *Current issues in ASL phonology: Current issues in ASL phonology* (pp. 227–261). Elsevier.

Prince, A., & Smolensky, P. (2004). Optimality in Grammar: Core Syllabification in Imdlawn Tashlhiyt Berber. In *Optimality Theory: Optimality Theory* (pp. 14–26). John Wiley & Sons, Ltd. https://doi.org/10.1002/9780470759400.ch2

Saffran, J. R., & Kirkham, N. Z. (2018). Infant statistical learning. *Annual Review of Psychology*, *69*, 181–203.

Saffran, J. R., Aslin, R. N., & Newport, E. L. (1996). Statistical learning by 8-month-old infants. *Science*, *274*(5294), 1926–1928.

Sandler, W. (1993). A sonority cycle in American Sign Language. *Phonology*, *10*(2), 243–279.

Selkirk, E. (1984). On the major class features and syllable theory. *Language Sound Structure*.

Speer, R. (2022, September). *rspeer/wordfreq: v3.0*. Zenodo. https://doi.org/10.5281/zenodo.7199437

Stone, A., Petitto, L.-A., & Bosworth, R. (2018). Visual sonority modulates infants' attraction to sign language. *Language Learning and Development*, *14*(2), 130–148.

Sun, C., & Billington, R. (2024). Sonority patterns in Lelepa onset clusters. *Proceedings of the Nineteenth Australasian International Conference on Speech Science and Technology*, 17–21.

Voronov, M., & Forkel, R. (2019, June). *OneAdder/lingtypology: Release 0.8.5*. Zenodo. https://doi.org/10.5281/zenodo.3247007

Yin, R., Weijer, J. van de, & Round, E. R. (2023). Frequent violation of the sonority sequencing principle in hundreds of languages: how often and by which sequences?. *Linguistic Typology*, *27*(2), 381–403.